Quiz 4

Graded

Student

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Total Points

100 / 100 pts

Question 1

(no title) — **20** / 20 pts

✔ **+ 20 pts** Correct

+ 0 pts Incorrect

Question 2

(no title) — **20** / 20 pts

✔ **+ 20 pts** Correct

+ 0 pts Incorrect

Question 3

(no title) — **20** / 20 pts

✔ **+ 20 pts** Correct

+ 0 pts Incorrect

Question 4

(no title) — **20** / 20 pts

✔ **+ 20 pts** Correct

+ 0 pts Incorrect

Question 5

(no title) — **20** / 20 pts

✔ **+ 20 pts** Correct

+ 0 pts Incorrect

Q1
20 Points

In RLHF, the Bradley-Terry model is used to model preferences. Which of the following best describes its application?

- ○ It predicts the most likely sequence of actions based on user ratings.
- ◉ It assigns probabilities to outcomes based on pairwise preferences.
- ○ It optimizes the policy directly using rewards.
- ○ It penalizes outputs that deviate from a reference distribution.

Q2
20 Points

A PPO model updates its policy using the Clip-PPO objective, with a trust region constraint $\epsilon = 0.2$.
Given the following probabilities:

Current policy: $\pi_\theta(a|s) = 0.7$
Old policy: $\pi_{\theta_k}(a|s) = 0.5$
Advantage estimate: $A^{\pi_{\theta_k}}(s, a) = 2.0$

What is the final loss $L(s, a, \theta_k, \theta)$ from this update?

Hint:
The policy ratio is defined as: $r(\theta) = \frac{\pi_\theta(a|s)}{\pi_{\theta_k}(a|s)}$

and the clipped ratio used in the update is: $\min(\max(r(\theta), 1-\epsilon), 1+\epsilon)$.

2.4

Q3
20 Points

A model is trained using pairwise comparisons in the Bradley-Terry framework. Two outputs A and B have logits:

- $\text{logit}_A = 4$
- $\text{logit}_B = 1$

Using the logits, compute the probability that A is preferred over B.
Hint: Use the softmax equation: $P(A) = \frac{e^{\text{logit}_A}}{e^{\text{logit}_A}+e^{\text{logit}_B}}$
Round your answer into three decimal points.

0.953

Q4
20 Points

How does supervised fine-tuning (SFT) affect a language model?

- ○ It guarantees perfect generalization to all unseen tasks without any trade-offs
- ◉ It improves performance on labeled examples
- ○ It eliminates the need for any human-provided data during training
- ○ It only changes the model's architecture, not its behavior

Q5
20 Points

Which statement best describes the key difference between Supervised Fine-Tuning (SFT) and Reinforcement Learning from Human Feedback (RLHF)?

- ◉ SFT trains on labeled input-output pairs, while RLHF optimizes model behavior based on reward signals from human preferences
- ○ SFT uses reward models, while RLHF only uses fixed labeled datasets
- ○ SFT does not require human annotation
- ○ RLHF does not require human annotation